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SUBJECT CODE NO: - RNP-8064
FACULTY OF SCIENCE AND TECHNOLOGY
S.E (EEE / EE / EEP) (NEP) Sem - IV
Examination May/June 2026
Network Analysis

[Time: 1:00 Hour]

[Max. Marks: 30]

Please check whether you have got the right question paper.

N. B

- 1) Part A: Question No 1 is compulsory.
- 2) Attempt any four questions from remaining Part B: Q.2 to Q.7.

Part A

Q.1 Solve any 5 following questions.

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- 1) Z-parameters are also called:
A) Impedance Parameters B) Admittance Parameters
C) Hybrid Parameters D) Transmission Parameters
- 2) A tree in graph theory contains:
A) All loops B) No loops
C) Only nodes D) Only branches
- 3) At resonance in a series RLC circuit:
A) $X_L > X_C$ B) $X_C > X_L$
C) $X_L = X_C$ D) $X_L = 0$
- 4) Node analysis is based on:
A) KVL B) KCL
C) Ohm's Law D) Superposition
- 5) Thevenin's theorem replaces a network by:
A) Current Source
B) Voltage Source in Series with Resistance
C) Only Resistance
D) Only Voltage
- 6) Maximum power transfer occurs when:
A) $R_L = 0$ B) $R_L = R_{th}$
C) $R_L = 2R_{th}$ D) $R_L = R_{th}/2$
- 7) Y- Parameters are also known as:
A) Impedance Parameter B) Admittance Parameter
C) Reactance Parameter D) None of the above
- 8) Time constant of an RC circuit is:
A) R/C B) RC C) $1/RC$ D) R^2C

Part B

Solve any four from the following

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| Q.2 Explain Thevenin's theorem with procedure. | 05 |
| Q.3 State and explain superposition theorem. | 05 |
| Q.4 Define graph, tree, and co-tree in network theory. | 05 |
| Q.5 Explain symmetrical and asymmetrical networks. | 05 |
| Q.6 Compare series resonance and parallel resonance (impedance, current, power factor. | 05 |
| Q.7 Explain two-port network Z parameters. | 05 |